

Chapter Three

The Parallel Multi-Dimensional Syntactic Building Hypothesis

In the preceding chapter, we introduced Merge and the theoretical apparatus associated with it. In the present chapter, we return to what may be regarded as one of the earliest questions of syntactic inquiry: how is a sentence built? Readers may already have noticed a small but telling terminological preference in this book: the term *generation* is used only sparingly, whereas *building* is repeatedly foregrounded. The reason is not merely stylistic. Human beings are not born with knowledge of any particular language, that is, of any determinate native language. What exists, rather, is an initially rudimentary language faculty, which gradually develops and matures, enabling the acquisition of specific languages one by one. It is precisely this gradual growth and maturation that may more properly be called *generation*. When we analyse word-formation and sentence-building, by contrast, what is at issue is a somewhat different process: once elements in the syntactic terminal list have been inscribed, they are built into structures, which are then handed over to the lexical list and encyclopaedic knowledge for the optimisation of underdetermined valuation, before receiving their eventual interpretation. For this reason, the term *building* is deployed here to refer to syntactic and lexical operations of construction, rather than *generation*.

Returning, then, to the question itself: to ask how a sentence is built is, in rather simple terms, to ask how a number of elements — namely, the features discussed in the previous chapter — are taken from the relevant inventory, placed within a workspace, and subjected to free Merge. In this chapter, I shall approach this question by developing an account of parallel multi-dimensional syntactic building.

3.1 From the Single Syntactic Tree to Parallel Multi-Dimensional Building

Since the emergence of syntactic analysis, attention has tended to fall upon the structure immediately before us: the sentence under analysis. It is therefore unsurprising that the analytic gaze has often remained local, even narrowly so. Within the Minimalist Program, as noted in the previous chapter, an especially salient point in early Minimalism — Chomsky (1993, 1995), together with earlier Chomskyan theories of formal linguistics — is that Merge, whether External Merge or Internal Merge, was commonly taken to be purpose-driven. External Merge was understood as satisfying selectional features, while Internal Merge was motivated by some further

requirement, such as feature-checking. From Chomsky (2004) onwards, however, External Merge and Internal Merge were unified under the single operation Merge, and Chomsky began to suggest that Merge is free. This line of thought was developed in Chomsky's later formulation of Simplest Merge, especially in Chomsky (2013, 2015), namely POP and POP+. Free Merge there receives a more explicit theoretical articulation: Chomsky (2015) assumes that "operations can be free, with the outcome evaluated at the phase level for transfer and interpretation at the interfaces."

While studying and reading works on the Minimalist Program, I encountered what seems to me a rather interesting problem. Once the relevant inscriptions have entered the workspace, why should the first two elements selected for Merge so reliably build the most basic structure in the appropriate way? How, from the workspace, do we "choose" the core of the propositional domain, namely the head *v* and its complement? Yet if such a "choice" is admitted, it appears to sit uneasily with the thesis of free Merge. What must therefore be explained is how the initial act of syntactic building is launched. It is against this background that I propose the Parallel Multi-Dimensional Syntactic Building Hypothesis.

3.2 Definition of the Parallel Multi-Dimensional Syntactic Building Hypothesis

On the basis of the preceding discussion, this book proposes the Parallel Multi-Dimensional Syntactic Building Hypothesis. Its central claim is that syntactic building does not, from the outset, proceed along a single correct derivational path. Rather, on the basis of free Merge, it gives rise to multiple derivational dimensions that run in parallel. By way of metaphor, "parallel multi-dimensionality" may be likened to a set of parallel universes: initial Merge opens, as it were, a multi-timeline space of syntactic building. All pairwise free-Merge structures formed from the elements in the workspace coexist in parallel across distinct dimensions. Each dimension has its own stage of construction — that is, its own phase — and upon completion each such stage is transferred to the interfaces for interpretation. The syntactic structure that we ordinarily analyse, then, is in fact one of the outcomes of parallel multi-dimensional syntactic building that has converged after interface evaluation. Sentences whose syntactic building structures are identical but whose meanings are ambiguous may likewise be understood as two distinct interpretable results emerging in different dimensions of this parallel multi-dimensional space.

The "dimension" at stake here is not a spatial dimension in the physical sense, nor need it be understood as a set of multiple computational processes literally unfolding simultaneously in the brain. More precisely, a dimension refers to a possible syntactic derivational path opened by free Merge. If there are several syntactic objects in the workspace — that is, elements inscribed from the syntactic terminal list — then, in principle, any two of them may enter into Merge. Each

possible instance of first Merge generates a new syntactic object and, at the same time, opens a possible derivational dimension. No distinction between “correct” and “incorrect” is stipulated in advance across these dimensions, since in a system of free Merge, Merge itself carries no inherent purposiveness and does not presuppose the satisfaction of some predetermined feature. Merge merely builds syntactic structure; whether that structure can be interpreted is determined subsequently, at the level of phase formation, by the interfaces. The notion of “dimension” adopted here also differs from the “derivational dimension” proposed by Pan and Du (2024). Their dimension is a derivational dimension driven by Pair-Merge and Set-Merge, whereas the “parallel multi-dimensionality” proposed in this book refers to a candidate derivational space generated by free Merge within the workspace. Put differently, Pan and Du’s dimension concerns the multi-planar architecture internal to a derivational result, while the present notion concerns the multiple derivational paths that free Merge may open. To avoid terminological confusion, this book will refer to Pan and Du’s derivational dimension as “derivational dimension” in the narrower sense. Their account can, of course, be traced back to Chomsky’s discussion of Pair-Merge; they also point out that the idea of a separate plane reflects a more general hypothesis, namely that human language contains more than one dimension of derivation.

A further distinction must nevertheless be drawn between “derivational dimension” and “parallel building dimension”. The separate plane of Chomsky (2019) and the derivational dimension of Pan and Du (2024) primarily explain a particular kind of dimensional space within a parallel building path: namely, which elements within that dimensional space occupy the same abstract computational plane, and which occupy different abstract computational planes. In adjunct structures, for instance, an adjunct may, through Pair-Merge, stand in a different derivational dimension from the main structure. Each derivational dimension has its own labelling algorithm, and the labelling algorithm operating within one dimension ignores syntactic objects located in other derivational dimensions. In this way, the adjunct does not interfere with the labelling of the main structure, and the relevant structure is then sent to the C-I interface for interpretation.

By contrast, the “parallel building dimension” proposed in this book addresses a more front-end problem. If Merge is free, how, at the initial stage of syntactic building, are the objects for initial Merge determined among the multiple syntactic objects in the workspace? In other words, the derivational dimension of Pan and Du (2024) is chiefly concerned with the multi-planarity internal to syntactic structure, that is, the separate-plane problem raised by Chomsky (2019). The parallel building dimension proposed here, however, concerns the multi-linearity of derivational paths. The former asks how components internal to a complex syntactic object are distributed across different computational planes; the latter asks how free Merge opens multiple candidate derivational paths from the workspace, and how interface evaluation ultimately filters these paths so that convergent structures remain.

Thus, Chomsky's (2019) idea of the separate plane and Pan and Du's (2024) model of derivational dimensions may be regarded as significant theoretical sources for the Parallel Multi-Dimensional Syntactic Building Hypothesis advanced here. Taken together, they suggest that syntactic computation need not be confined to a single plane. Building on this insight, the present proposal argues that syntactic computation may display multi-planarity not only in adjuncts, coordinate structures, and multiple modifiers, but also multi-pathicity at the initial stage of free Merge. Put differently, if theories of derivational dimension tell us that there may be multiple computational planes inside syntactic structure, the present hypothesis proposes that syntactic building itself may unfold from multiple candidate derivational routes. The former supports the latter, but the latter takes one step further. The former chiefly explains adjuncts, coordinate structures, and labelling problems; the latter seeks to explain how, under free Merge, initial structures arise, compete, undergo filtering, and eventually converge.

On this basis, this book treats the Parallel Multi-Dimensional Syntactic Building Hypothesis as its central theoretical assumption. In order to render the hypothesis more explicit, we may offer a preliminary formalisation.

Suppose that several syntactic objects have been inscribed into the workspace. Under the condition of free Merge, any two objects x and y in the workspace may, in principle, undergo Merge:

$$\text{Merge}(x, y) = \{x, y\}$$

Consequently, all possible instances of initial Merge in that workspace may form a set of candidate Merge operations:

$$M(\text{WS}) = \{ \text{Merge}(x, y) \mid x, y \in \text{WS}, x \neq y \}$$

That is, so long as x and y both belong to the current workspace, and x is not identical to y , they may in principle enter into Merge. If the workspace contains n syntactic objects, then, setting aside linear order, the number of candidate derivational routes opened by initial candidate Merge is:

$$n(n - 1) / 2$$

Each possible instance of initial Merge opens a candidate derivational route. These routes are, in theoretical terms, parallel to one another and may be called different parallel building dimensions. The set of parallel building dimensions generated by the workspace WS may therefore be represented as follows:

$$D(\text{WS}) = \{ dxy \mid dxy = \text{Merge}(x, y), x, y \in \text{WS}, x \neq y \}$$

If the syntactic objects α , β , γ , and δ are inscribed in the workspace, such that:

$$WS = \{\alpha, \beta, \gamma, \delta\}$$

then the following candidate derivational dimensions may be generated:

$$\begin{aligned} d1 &= \text{Merge}(\alpha, \beta) \\ d2 &= \text{Merge}(\alpha, \gamma) \\ d3 &= \text{Merge}(\alpha, \delta) \\ d4 &= \text{Merge}(\beta, \gamma) \\ d5 &= \text{Merge}(\beta, \delta) \\ d6 &= \text{Merge}(\gamma, \delta) \end{aligned}$$

Here, $d1$, $d2$, $d3$, and so forth designate different candidate derivational dimensions. At this point, they are parallel and mutually independent. Nor can they yet be judged correct or incorrect. Under a theory of free Merge, Merge itself is responsible only for building syntactic structure; it is not responsible for judging in advance whether that structure will ultimately converge. When a given candidate dimension continues to build and forms a staged syntactic unit, that is, a phase Ph, the candidate dimension enters interface evaluation. We may represent the phase-level evaluation function as EvalPh:

$$\begin{aligned} \text{EvalPh}(d_i) = 1 &\rightarrow d_i \text{ may be transferred and continue to converge} \\ \text{EvalPh}(d_i) = 0 &\rightarrow d_i \text{ fails at the interface and is eliminated} \end{aligned}$$

EvalPh is not a selection mechanism prior to Merge, but an evaluation mechanism after Merge. Different dimensions are not predetermined as correct or incorrect before Merge; rather, after subsequent phase formation, they are evaluated at the interface level. Dimensions that can be interpreted by the C-I interface and the SM interface converge, whereas those that cannot be interpreted are eliminated, as in (1).

$$(1) \text{ EvalPh}(d_i) = 1 \text{ iff } CI(d_i) = 1 \text{ and } SM(d_i) = 1$$

Here, $CI(d_i) = 1$ means that d_i can receive an interpretation at the conceptual-intentional interface; $SM(d_i) = 1$ means that d_i can be externalised at the sensory-motor interface. A candidate dimension can therefore converge only if it satisfies at least the condition in (2).

$$(2) \text{ Converge}(d_i) \text{ iff } \text{EvalPh}(d_i) = 1 \text{ at every relevant phase Ph}$$

In other words, a candidate dimension may become a convergent derivation only if it passes interface evaluation at every relevant phase level. If a given dimension fails interface evaluation at any crucial phase level, that dimension is eliminated, as shown in (3).

$$(3) \text{ Crash}(d_i) \text{ iff } \exists \text{ Ph such that } \text{EvalPh}(d_i) = 0$$

Given an initial workspace WS, the set of dimensions that can ultimately converge may be represented as in (4).

$$(4) \text{Conv(WS)} = \{ d_i \in D(\text{WS}) \mid d_i \text{ passes EvalPh at every relevant phase} \}$$

The syntactic tree we ordinarily draw in syntactic analysis may be understood as one member of Conv(WS) :

$$T = \text{one member of } \text{Conv(WS)}$$

If Conv(WS) contains only one member, the sentence has only one convergent structure. If Conv(WS) contains multiple members, and if those members correspond to different interface interpretations, structural or semantic ambiguity may arise, as in (5).

$$(5) \text{ If } |\text{Conv(WS)}| > 1, \text{ then ambiguity may arise.}$$

Certain sentences, for instance, may have only one phonological output at the surface while permitting two or more semantic interpretations. This can be understood as a case in which multiple parallel building dimensions receive legitimate interpretations at the interfaces simultaneously. Since none of them is eliminated at the EvalPh stage, all may enter the final interpretive space. The Parallel Multi-Dimensional Syntactic Building Hypothesis may therefore be summarised as in (6).

(6) The Parallel Multi-Dimensional Syntactic Building Hypothesis

Given a workspace WS, if Merge is free, then any two distinct syntactic objects in WS may undergo Merge. Each possible instance of Merge generates a candidate derivational dimension. The derivational paths of different candidate dimensions unfold in parallel and undergo interface evaluation after phase derivation has been completed within their respective dimensions. Dimensions that pass interface evaluation converge; those that fail interface evaluation crash. The structure ultimately analysed, then, is not a unique structure preset before Merge, but one of the legitimate results preserved after interface filtering within the space of parallel building dimensions.

The Parallel Multi-Dimensional Syntactic Building Hypothesis may be condensed into four claims: (a) the elements in the workspace may, in theoretical terms, undergo free Merge; (b) different modes of first Merge form different possible building routes, namely different “parallel dimensions”; (c) only those dimensions that can receive interpretation at the interfaces may ultimately become legitimate syntactic structures; and (d) each dimension is derived independently. Once each derivational stage is completed, it is transferred and evaluated at the interfaces. If it

passes, building proceeds to the next stage; if it fails, it crashes. This continues until the building operation has exhausted the elements in the workspace.

3.3 Parallel Dimensions and Systemic Simplicity

In the preceding section, I proposed the hypothesis of Parallel Multi-Dimensional Syntactic Building. A question arises almost immediately. If, once such multi-dimensional derivation is set in motion, the system is expected to operate over all the elements in the workspace, thereby yielding multiple derivational paths unfolding in parallel, does syntax not become complex from the very beginning? Does this hypothesis not stand in tension with the simplicity pursued by the Minimalist Programme?

This is precisely the sort of question that any new hypothesis ought to face. Simplicity, as one of the central aims of syntactic theory after the advent of the Minimalist Programme, consists in explaining the complexity of human language with as few mechanisms as possible. At first sight, the present hypothesis may seem to move in the opposite direction. Since it allows for a considerable number of parallel derivational dimensions, it may appear to add complexity rather than reduce it. Yet this concern, I would suggest, rests on a conflation of two distinct notions: the complexity of the operation itself and the complexity of the candidate space. It is the former, rather than the latter, that is most directly relevant to minimalist simplicity.

By the complexity of an operation, I mean the extent to which the syntactic system is required to undertake additional tasks in the construction of a sentence. Within the Minimalist Programme, Merge is taken to be the sole, or at least the most fundamental, operation of syntactic building. If, however, Merge is understood as a purpose-driven operation in the sense discussed above—if some regulatory device in the workspace is assumed to determine in advance which two syntactic objects should merge, which derivational path will eventually converge, and which structure will be legible at the interfaces—then Merge is simple only in appearance. It would not merely combine objects. It would also have to anticipate structure, identify the relevant target, and predict interpretive outcomes. Burdened with such teleological modules, Merge would become a considerably more complex operation than its minimalist formulation would otherwise suggest.

Precisely here lies the tension with the basic spirit of free Merge. If Merge is free, it should not already possess the “right answer” before the operation has taken place. Merge itself is responsible only for combining two syntactic objects. Whether the resulting object can form a legitimate structure, receive a semantic interpretation, or suit a particular discourse context should be assessed subsequently, at the level of phases and interfaces. Put differently, the syntactic system need not select a single correct path before Merge occurs. It may instead

generate multiple candidate derivational dimensions on the basis of free Merge, leaving the interface systems to filter them. This filtering process is dynamic; in a relevant sense, it is also non-retrospective. The Parallel Multi-Dimensional Syntactic Building Hypothesis may therefore be understood as a way of avoiding operational complexity. In line with other minimalist approaches, it deploys only one operation of sentence construction, namely Merge, while still assigning a salient role to the interfaces.

This understanding is not merely a theoretical fancy. Ordinary language use itself points in a similar direction. In real-time speech production, speakers do not always construct sentences along a fully determinate and wholly predictable route from the outset. Sentence construction may be regarded, at least in part, as the compositional assembly of the meaning one intends to express. Yet such an assembly may give rise to utterances that are semantically possible but pragmatically infelicitous. We often notice, after having spoken, that what we have said does not quite fit the occasion; or, while still speaking, we may sense that the next stretch of the utterance is becoming inaccurate, inappropriate, or ill-suited to the intended context, and so revise it mid-course. Such facts suggest that sentence construction is not wholly predictable. They also point to one respect in which human linguistic production differs from machine-generated language: it is not always reducible to a single, pre-specified line of derivation.

This does not require us to claim that the brain literally and exhaustively derives every possible syntactic structure. A more cautious formulation, and one more consonant with the hypothesis developed here, would be this: human language production involves candidacy, competition, and revision. During syntactic building, several possible expressive paths may be activated to different degrees. Some are allowed to proceed and may ultimately be externalised. Others fail to receive an appropriate interpretation at the interfaces and are abandoned. Still others may be interpretable at one level, yet later discarded on semantic, pragmatic, or externalisational grounds. A given path may even be relinquished rather abruptly at a particular stage of derivation. This picture is consistent with the Parallel Multi-Dimensional Syntactic Building Hypothesis proposed in this book. The earlier metaphor of “parallel universes”, then, should not be dismissed as a merely decorative figure of speech. It may be understood, at least minimally, as the candidate derivational space synchronically opened by initial Merge under a theory of free Merge. Different dimensions coexist in parallel during syntactic construction, compete with one another, and are ultimately selected, or rejected, by the interfaces for expression.

The hypothesis thus has a certain theoretical and operational simplicity. Like the more persuasive versions of minimalist theory, it does not place excessive predictive machinery inside Merge itself. It requires Merge to do only its most elementary work: to combine two elements into a new syntactic object. If free Merge is allowed to generate candidate structures, and if the interfaces are then

responsible for interpreting and filtering those structures, Merge is kept in its simplest form. What appears to be a complex process is not built into the operation of Merge itself. Rather, it is distributed across the competition among candidate dimensions, phase-level evaluation, and interface interpretation.

It should also be made clear that the “parallel multi-dimensionality” proposed here does not mean that syntax may generate structures without constraint, nor that every possible syntactic structure can develop into a sentence. The claim is more modest. Within the framework of free Merge, syntactic building may begin from multiple candidate paths. The sentence that we eventually utter, hear, and analyse is only one convergent result that has survived interface filtering. In this sense, the metaphor of “parallel universes” acquires a more serious interpretation. It refers not to multiple universes in the physical world, but to the parallel opening of multiple possible derivational paths within syntactic computation.

3.4 Prospects and Open Questions

For a textbook, one may reasonably ask why a chapter of this kind is needed at all: why introduce a new theoretical hypothesis, especially one that does not directly alter the basic presentation of syntactic analysis? My answer is that the hypothesis offers a fresh way of understanding free Merge. It also provides what seems to me a promising account of why initial Merge so often appears to instantiate a head–complement relation. At the same time, one should be careful not to overstate the case. The Parallel Multi-Dimensional Syntactic Building Hypothesis remains a preliminary theoretical framework. If it is to become a mature syntactic model, several open questions must still be addressed.

One such question concerns psychological reality. Can the human brain genuinely carry out multiple derivations of sentence structure at the same time? This is not a matter that syntactic theory alone can settle. It will require further work in neurolinguistics and psycholinguistics. Still, the point should be restated with some caution: candidate derivational dimensions in grammatical theory need not be identical to fully simultaneous derivational computations in the brain. Whether parallel dimensions possess psychological reality, and in what sense they might do so, remains an open empirical question.

A second issue concerns limitation. Multi-dimensionality does not imply unrestricted expansion. The pivotal question is how the relevant limiting mechanisms should be formalised. Third-factor principles, among others, may play a role here. This matter will be taken up again in the following chapters.

Two further questions are especially salient: empirical prediction and integration with the internal mechanisms of the Minimalist Programme. The first asks whether the hypothesis can help us reinterpret concrete linguistic phenomena,

and whether such reinterpretation has genuine explanatory value. I suspect that ambiguity will be a particularly fruitful domain in which to pursue this line of enquiry. The second question concerns theoretical integration. The hypothesis proposed here is not intended to replace the architecture of the Minimalist Programme, but to supplement it. In later chapters, I shall consider how it may be brought into relation with other minimalist mechanisms.

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